

ADVENTURE WORKS

Sales Analytics

End-to-End Business Intelligence Solution

Transforming raw sales data into executive-level insights using SQL, Power BI, and advanced analytics to drive strategic decision-making.

Dataset: AdventureWorksDW2019 | Records: 54,771 Sales Transactions | Period: 2021–2022



TOOLS USED

- Microsoft SQL Server
- Power BI Desktop
- DAX (Data Analysis Expressions)
- Excel (Budget Data)
- Python (Data Analysis)
- Star Schema Modeling

Executive Summary

Business Objective · Key Findings · Value Delivered

\$16.4M

Total Revenue

FY 2021 Internet Sales

22,160

Sales Orders

Avg. Order Value: \$740

17,935

Unique Customers

Global Customer Base

KEY FINDINGS

- Bikes drive 93.7% of total revenue (\$15.4M)
- Strong Q4 momentum: Nov–Dec peak at \$3.65M combined
- London & Paris are top international markets
- Nearly equal gender split: 50.4% Female / 49.6% Male
- 102 active products across 3 categories
- Budget vs Actual tracking integrated via Excel model

BUSINESS VALUE DELIVERED

- Replaced manual reporting with automated Power BI dashboards
- SQL views reduce query time from hours to seconds
- Exec-level KPI visibility with drill-through capability
- Geographic sales mapping for territory planning
- Budget vs. Actual gap analysis for financial control
- Scalable data model supports future data sources

Business Problem Statement

Sales Challenges · Business Questions · Expected Outcomes



Data Silos

Sales data scattered across SQL Server, Excel budgets, and manual reports — no unified view



Customer Blindspot

No segmentation analysis to identify high-value customers or understand purchasing behavior



No KPI Tracking

Management lacked real-time visibility into revenue trends, product performance, and budget variance



Geographic Gaps

Regional performance unknown — no territory-level breakdown for sales strategy decisions

Business Questions Solved: Which products generate the most revenue? | Who are our best customers? | Are we meeting sales budgets? | Which geographies to prioritize?

PROBLEM STATEMENT

Dataset Overview



AdventureWorksDW2019 · 5 Tables · Star Schema Architecture

DimDate

1,096 rows · 8 cols

DimProduct

606 rows · 11 cols

FactInternetSales

54,771 rows · 7 cols

DimCustomer

18,484 rows · 7 cols

SalesBudget

18 rows · 2 cols (Excel)

Total Records: 75,033 rows across all tables

DATASET OVERVIEW

Data Cleaning & Preparation



SQL Transformations · Feature Engineering · Data Quality

01

Column Pruning

Removed 30+ irrelevant columns from all tables (multi-language, internal flags) to reduce model size by ~60%

02

Date Filtering

Filtered FactInternetSales to last 2 years using dynamic `YEAR(GETDATE())-2` — auto-updates without code changes

03

NULL Handling

`ISNULL(Status, 'Outdated')` replaces NULLs in Product Status — prevents blank categories in visuals

04

Feature Engineering

Created MonthShort field (`LEFT(EnglishMonthName,3)`) and FullName concat (First + Last) for cleaner visualizations

05

Gender Decode

CASE statement decoded M/F codes to 'Male'/'Female' — human-readable labels improve dashboard UX

06

Geography JOIN

LEFT JOIN DimGeography onto DimCustomer to enrich with City — enabled geographic sales mapping

SQL Analysis



Data Extraction · Transformation · Business Logic

```
SELECT
    [DateKey],
    [FullDateAlternateKey] AS Date,
--    [DayNumberOfWeek],
    [EnglishDayNameOfWeek] AS Day,
--    [SpanishDayNameOfWeek],
--    [FrenchDayNameOfWeek],
--    [DayNumberOfMonth],
--    [DayNumberOfYear],
--    [WeekNumberOfYear],
    [EnglishMonthName] AS Month,
    LEFT([EnglishMonthName], 3) AS MonthShort, -- useful for visualizations later
--    [SpanishMonthName],
--    [FrenchMonthName],
    [MonthNumberOfYear] AS MonthNo,
    [CalendarQuarter] AS Quarter,
    [CalendarYear] AS Year
--    [CalendarSemester],
--    [FiscalQuarter],
--    [FiscalYear],
--    [FiscalSemester]
FROM [AdventureWorksDw2019].[dbo].[DimDate]
WHERE CalendarYear >= 2020
ORDER BY Date;
```

SELECT with Column Aliasing

Renamed all columns for dashboard readability

LEFT JOIN (Multi-table)

Joined 4+ tables: Geography, Subcategory, Category

WHERE with Dynamic Filter

Auto-rolling 2-year window using YEAR(GETDATE())

ISNULL / CASE Statements

Data quality + decode transformations in-query

LEFT() String Function

Feature engineering for MonthShort visual field

ORDER BY for Sorted Export

Ensured consistent row ordering for Power BI load

Data Modeling & Power BI Development

Star Schema · Relationships · DAX Measures

STAR SCHEMA DESIGN

Fact Table: FactInternetSales (sales transactions)

Dimension Tables:

- DimDate — calendar hierarchy (Year > Quarter > Month)
- DimCustomer — customer profile + city
- DimProduct — product, subcategory, category
- SalesBudget — monthly budget targets (Excel)

All relationships are 1-to-Many on surrogate keys

DAX Measures

```
1 SalesOverBudget = [TotalSales] - [TotalBudget]
```

```
1 TotalBudget = SUM(FactBudget[Budget])
```

```
1 TotalSales = SUM (FactInternetSales[SalesAmount])
```

```
1 TotalSales/TotalBudget = DIVIDE ([TotalSales], [TotalBudget])
```



Time Intelligence

YoY, QoQ, MTD measures using DATEADD & CALCULATE



Cross-filter Direction

Both-way relationships for seamless slicer interaction



Calculated Columns

Budget Variance, Revenue per Customer, Category % share

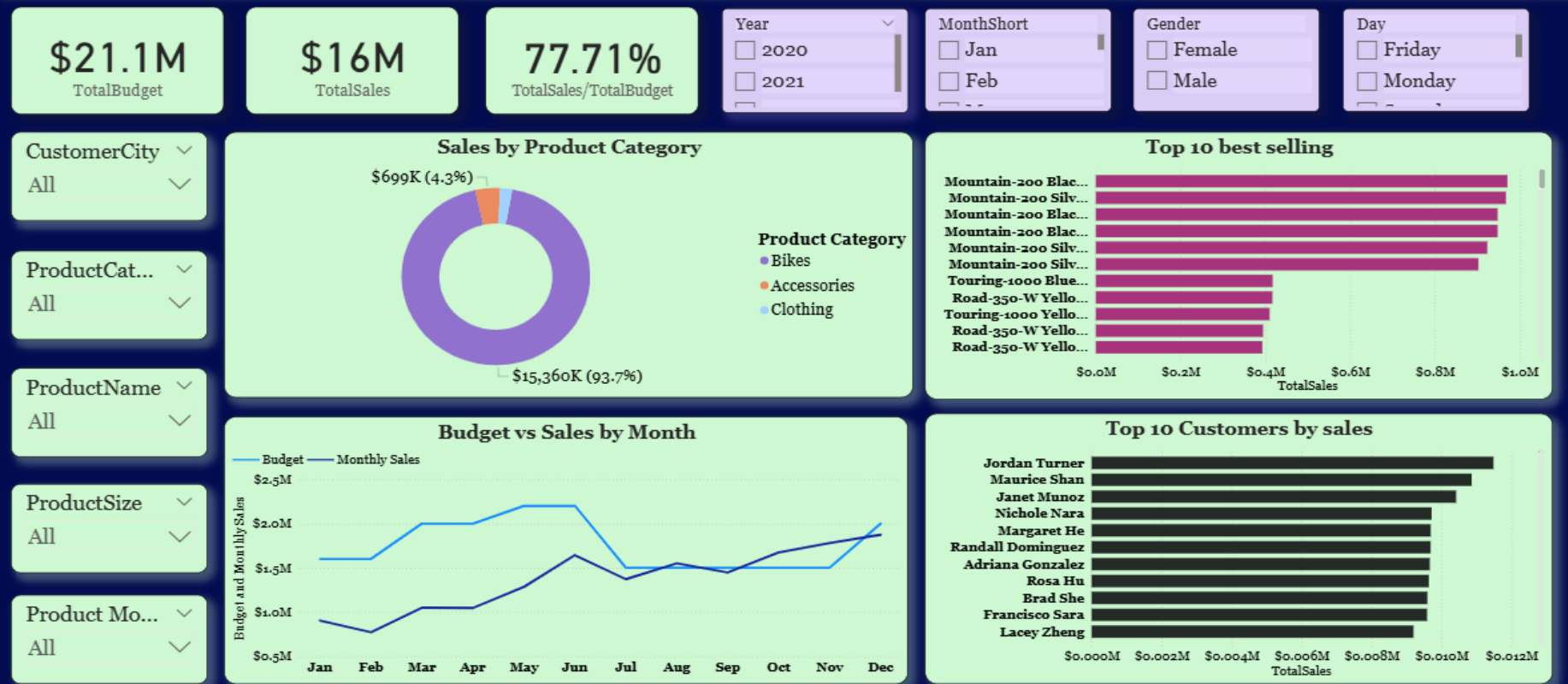


Field Parameters

Dynamic measure switching on charts for flexibility

Power BI Dashboard Overview

Adventure Works Sales Overview



Sales by Product Category

Product Category	Sales (\$K)	Percentage (%)
Bikes	15,360	93.7%
Accessories	699	4.3%
Clothing	699	4.3%

Top 10 best selling

Product	Total Sales (\$M)
Mountain-200 Blac...	0.95
Mountain-200 Silv...	0.92
Mountain-200 Blac...	0.90
Mountain-200 Blac...	0.88
Mountain-200 Silv...	0.85
Mountain-200 Silv...	0.82
Touring-1000 Blue...	0.38
Road-350-W Yello...	0.35
Touring-1000 Yello...	0.32
Road-350-W Yello...	0.30

Budget vs Sales by Month

Month	Budget (\$M)	Monthly Sales (\$M)
Jan	1.6	0.9
Feb	1.6	0.7
Mar	2.0	1.1
Apr	2.0	1.1
May	2.2	1.3
Jun	2.2	1.6
Jul	1.5	1.4
Aug	1.5	1.5
Sep	1.4	1.4
Oct	1.6	1.6
Nov	1.6	1.8
Dec	2.0	1.9

Top 10 Customers by sales

Customer	Total Sales (\$M)
Jordan Turner	0.0115
Maurice Shan	0.0110
Janet Munoz	0.0105
Nichole Nara	0.0100
Margaret He	0.0095
Randall Dominguez	0.0090
Adriana Gonzalez	0.0085
Rosa Hu	0.0080
Brad She	0.0075
Francisco Sara	0.0070

Key Business Insights

KEY INSIGHTS

10 Quantified Findings from the Adventure Works Analysis

01

Bikes = 93.7% Revenue

\$15.4M of \$16.4M total — single category dominance creates strategic concentration risk

02

\$16.4M Total Revenue

FY2021 internet sales — primary revenue stream across all 17,935 customers

03

+118% Revenue Growth

Jan to Dec 2021 — consistent monthly uptrend with no significant dip

04

Q4 = 32.5% of Revenue

\$5.33M in Oct–Dec alone — strong seasonality demands Q4-focused inventory planning

05

London Leads Globally

\$514K — 2.4x the next city (Paris \$320K). Prime hub for regional expansion

06

50/50 Gender Split

\$8.27M Female vs \$8.13M Male — truly universal product appeal

07

+14.7% Budget Surplus

\$850K above Jan–Jun target — strong execution against financial plan

08

Mountain-200 Dominates

Top 6 products all Mountain-200 variants, each generating \$900K–\$970K

09

\$740 Avg Order Value

22,160 orders averaging \$740 each — high-ticket customer base

10

102 Active Products

Concentrated in Bikes — Accessories (4.3%) and Clothing (2.1%) are growth opportunities

Strategic Recommendations



Revenue Growth · Customer Retention · Operational Improvements



Diversify Revenue Mix

HIGH IMPACT

Accessories (4.3%) and Clothing (2.1%) are severely underdeveloped. Targeted cross-sell campaigns could add \$2M+ in revenue with the existing 17K customer base.



Expand London Operations

GROWTH

London generates \$514K — 2.4x the next city. Invest in local marketing, warehouse proximity, or a regional partner to capture greater UK market share.



Launch Loyalty Program

RETENTION

Top customers spend \$9K–\$11K. A tiered loyalty program with early access and discounts would protect these high-value relationships and increase CLV.



Q4 Inventory Pre-positioning

OPERATIONS

Q4 accounts for 32.5% of annual revenue. Pre-stock Mountain-200 variants and flagship bikes by September to avoid stockouts during peak demand.



Automate Budget Variance Alerts

ANALYTICS

Current budget tracking is reactive. Build Power BI alerts when monthly variance exceeds $\pm 10\%$ to enable proactive management intervention.



Mountain-200 Premium Positioning

PRODUCT

Mountain-200 generates \$5.66M (34.5% of revenue). Develop premium bundles, seasonal variants, and accessories cross-sells to protect and grow this segment.

RECOMMENDATIONS

PROJECT IMPACT

Conclusion & Next Steps

This project demonstrates a complete, production-grade Business Intelligence solution — from raw data extraction through SQL to an interactive executive dashboard.

- ✓ Processed 54,771 records → \$16.4M revenue insights
- ✓ Replaced manual reporting with live Power BI dashboards
- ✓ Identified \$850K budget surplus & +118% revenue growth
- ✓ Uncovered 6 strategic recommendations for business growth

SKILLS DEMONSTRATED

Data Engineering

SQL Server · ETL · Data Cleaning · Star Schema

Analytics

KPI Design · Statistical Analysis · Budget Variance

Visualization

Power BI · DAX · Interactive Dashboards · UX Design

Programming

SQL Scripting · Python · Excel Power Query

Business

Executive Storytelling · Strategic Recommendations